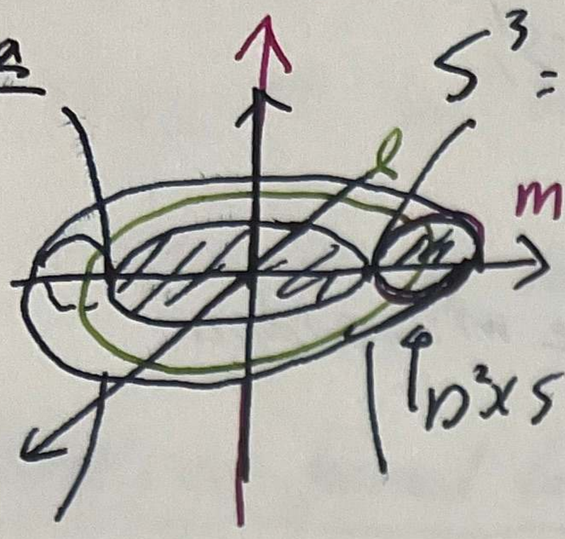
Exa



$S^3 = (D^2\times S^1) \cup_\varphi (D^2\times S^1) \longleftrightarrow S^3 = (S^3 - (D^2\times S^1)) \cup_\varphi D^2\times S^1$

$\varphi : \partial(D^2\times S^1) \to \partial(D^2\times S^1)$

$\| \qquad\qquad \|$

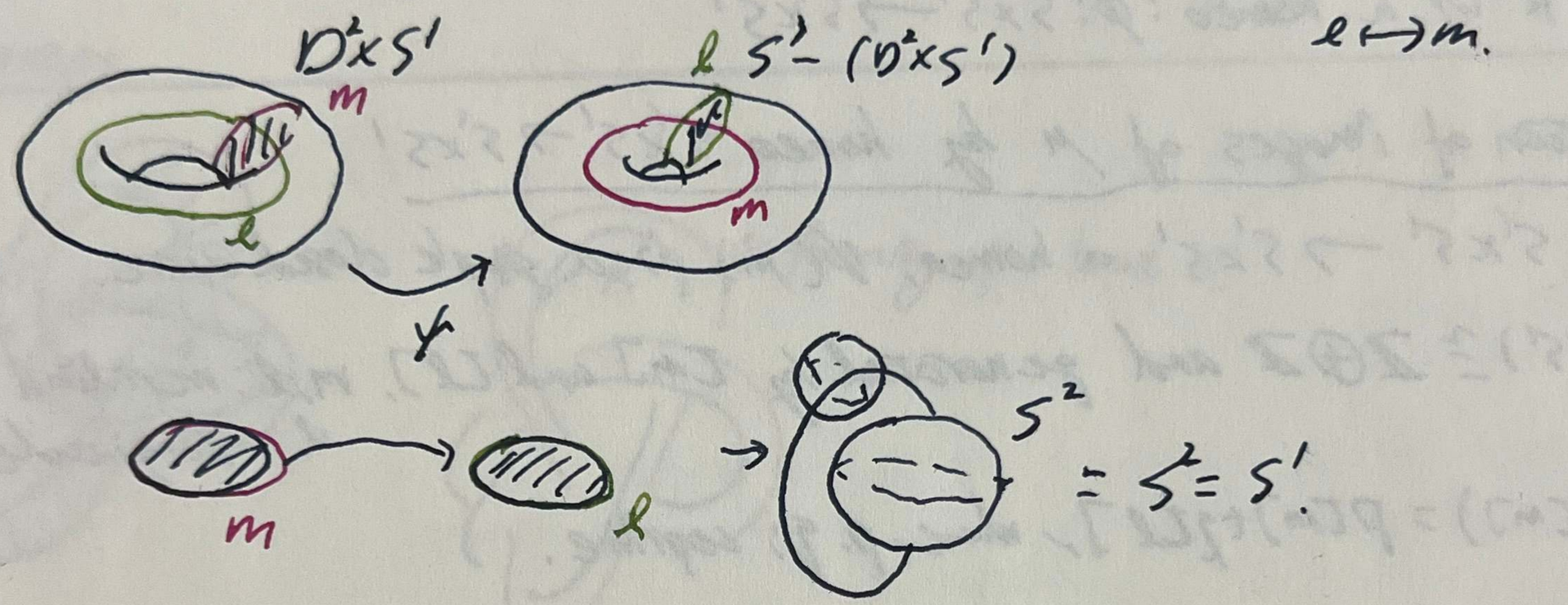
$S^1\times S^1 \to S^1\times S^1$

$m \longmapsto m$

$l \longmapsto l$

Note that S^3 can be obtained by attaching a solid torus to a torus boundary by a homeomorphism. This homeomorphism sends m to m l to l.

Exa Let us consider a homeomorphism $\psi : S^1\times S^1 \to S^1\times S^1$ s.t. $m \longmapsto l$, $l \longmapsto m$.



Here, in other words, we can obtain another 3-manifold & $S^2\times S^1$ by removing a solid torus and attaching solid torus again by a homeo.

$\psi : S^1\times S^1 \to S^1\times S^1$.

Note that the solid torus in the above examples can be considered as a regular neighborhood of a trivial knot. In general,

Def Let K be a knot. Let $N(K)$ be a regular neighborhood in S^3

$M := (S^3 - N(K)) \cup_\varphi (D^2\times S^1)$, $\varphi : S^1\times S^1 \to S^1\times S^1$: a homeo,

is a 3-manifold.

This process is called a Dehn surgery along a knot K on S^3.

M is called a 3-manifold obtained from S^3 by a Dehn surgery.

Thm $J_n(K,A) = (\not{*}((-1)^n A^{n^2+2n}))^{-wr(D)} \langle D^{S_n} \rangle$, n-th colored Jones poly

where S_n: Chebyshev poly in $S(\circlearrowleft) \cong \mathbb{Z}[A^{\pm 1}][x]$.

Reshetikhin-Turaev Invariant

Let D be a framed link diagram of n-components.

Let $\langle , \cdots, \rangle_D : S(\circlearrowleft \times I) \times \cdots \times S(\circlearrowleft \times I) \to S(\mathbb{R}^2)$

the threading operation.

Thm Supp. $A = e^{\frac{1}{4r}\pi i}$. i.e A is $4r$-th primitive root of unity. → $S_{r-1}=0, S_n=0 \; n \geq r-1$

$w = \sum_{n=0}^{r-2} \Delta_n S_n(x)$. $r \geq 3$.

If D' is obtained from D by second Kirby move, then

$$\langle w, \cdots, w \rangle_D = \langle w, \cdots, w \rangle_{D'}.$$

Thm Let L, L' be framed links & $M_L, M_{L'}$: corresponding 3-mfds.

$M_L \cong M_{L'}$: homeo $\Leftrightarrow$ L, L' are equiv. up to

① $\Omega_1', \Omega_2, \Omega_3$

② Kirby moves.

Revisit to Kauffman bracket and colored Jones polynomial

Kauffman bracket

$$\langle\ \rangle : \{\text{diagrams}\}/_{\Omega_1', \Omega_2, \Omega_3} \to \mathbb{Z}[A^{\pm 1}]$$

$$\langle \times \rangle = A \langle)(\rangle + A^{-1} \langle \smile\!\frown \rangle$$

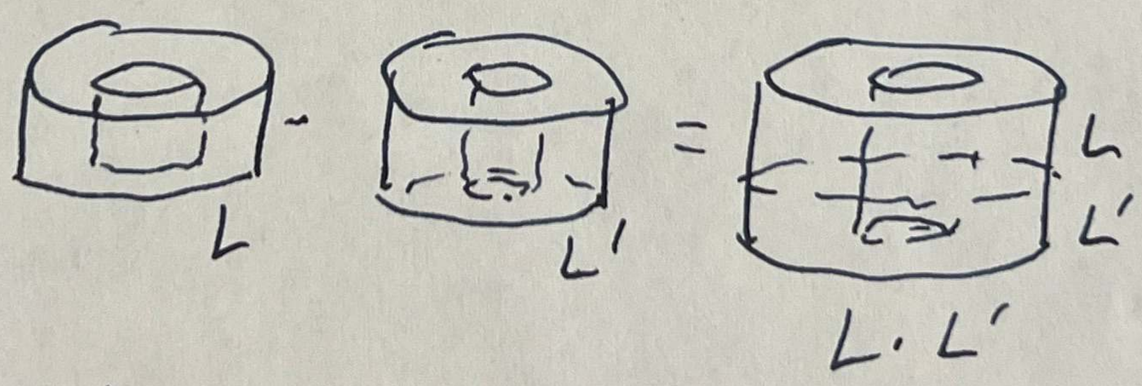
$$\langle \bigcirc \sqcup L \sqcup \bigcirc \rangle = (-A^2 - A^{-2}) \langle L \rangle.$$

Kauffman bracket skein algebra of $\Sigma \times I$

$S_{2,\infty}(\Sigma \times I)$:= a module over $\mathbb{Z}[A^{\pm 1}]$ generated by framed links in $\Sigma \times I$ quotiented by $\times = A)(+ A^{-1} \smile\!\frown$

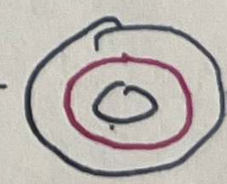
$\bigcirc \sqcup L = (-A^2 - A^{-2}) L$.

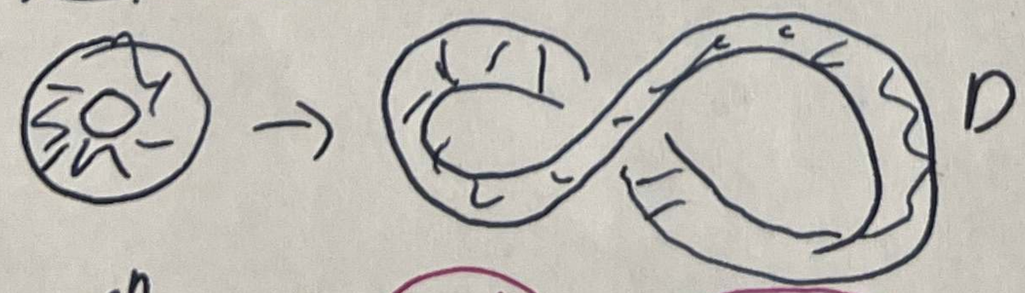
multiplication:

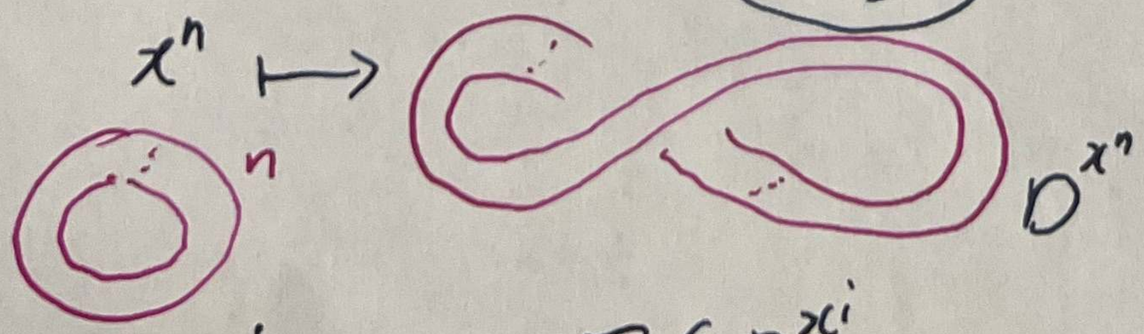


$(S_{2,\infty}(\Sigma \times I), \cdot)$ is called a Kauffman bracket skein algebra of $\Sigma \times I$.

Threading operation

$\because S(\bigcirc) \cong \mathbb{Z}[A^{\pm 1}][x]$, $x :=$ 

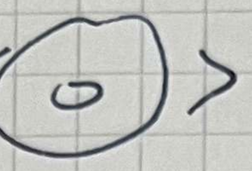




$p(x) = \sum c_i x^i \mapsto \sum c_i D^{x^i}$.

Remark skein algebra of  $\times I$.

S_n : Chebyshev poly.

Δ_n : $S_n \hookrightarrow$ 

Roseberg-Turaev invariant

Let $w = \sum_{n=0}^{r-2} \Delta_n S_n(x) \in S(\mathbb{R}^2 \times S^1)$, $r \geq 3$. D: a framed link diagram of 1-comp

$\Rightarrow \langle \cdot, \cdots, \cdot \rangle_D : S(\bigcirc \times I)^{\otimes n} \to S(\mathbb{R}^2)$: threading operation.

Thm Suppose that $A = e^{2\pi i \frac{k}{4r}}$. ie A is $4r$th root of unity.

If D' is obtained from D by second Kirby move, then

$$\langle w, \cdots, w \rangle_D = \langle w, \cdots, w \rangle_{D'}.$$

For the first Kirby move, we add or delete  and 

Let $N(K)$ (N_{ij}) where $N_{ij} = \begin{cases} lk(K_i, K_j) & i \neq j \\ w(K_i) & i = 0. \end{cases}$

Let $b_+(D)$ and $b_-(D)$ be the number of positive and negative eigenvalues of $N(K)$.

Thm Let $I(D) = \langle w, \cdots, w \rangle_D \langle w \rangle_{U_+}^{-b_+} \langle w \rangle_{U_-}^{-b_-}$.

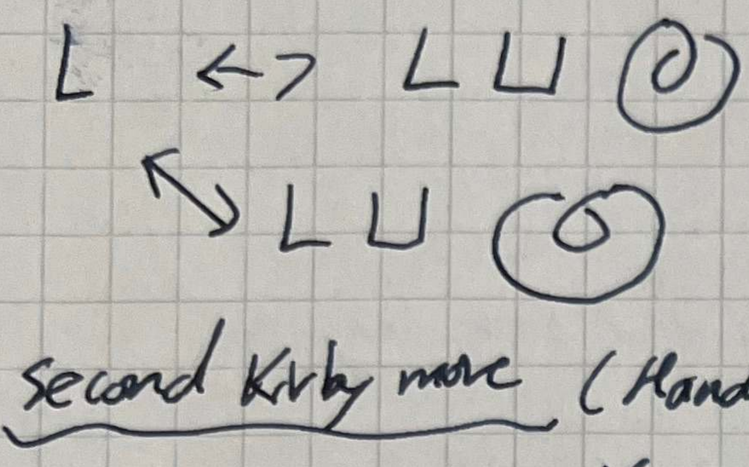
Then it is invariant under Kirby moves and Δ moves for framed links.

ie. It is topological invariant for any closed orientable 3-mfd

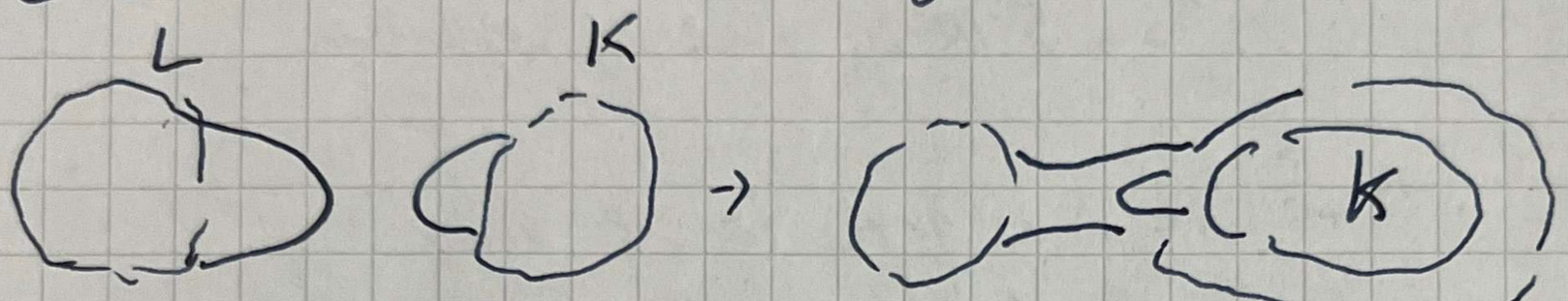
Kirby calculus

Let L be a framed link.

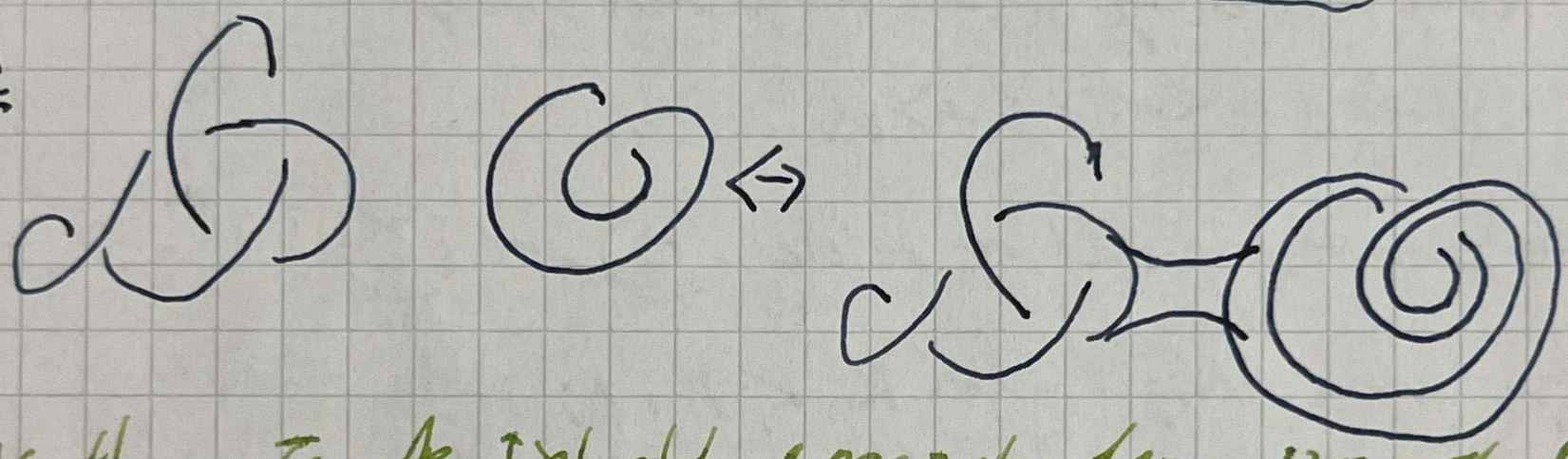
First Kirby move



Second Kirby move (Handle sliding move)



Ex



Kauffman - Temperley-Lieb algebra recently here. 13.5.2 thm ?

Thm Let L, L' be framed links. $M_L, M_{L'}$: corresponding 3-mfds.

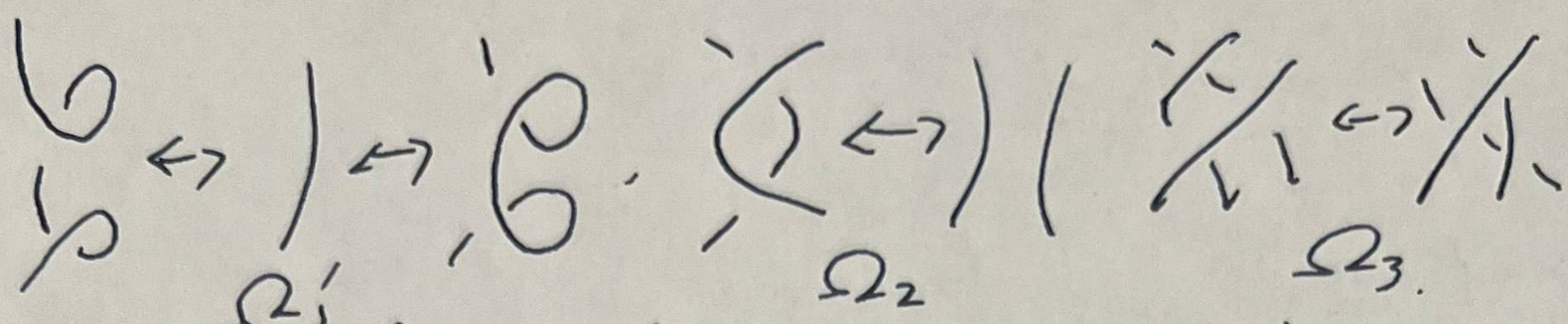
[Kirby] $M_L \cong M_{L'}$: homeo $\Leftrightarrow$ L, L' are equiv up to.

① as framed links

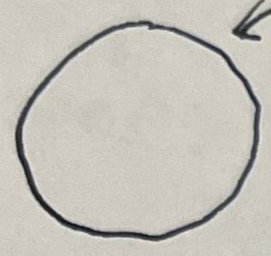
② Kirby moves.

Thm Let L, L' be framed link. $D_L, D_{L'}$: ori. diagrams

$L \sim L' \Leftrightarrow D_L$ and $D_{L'}$ are equiv. modulo

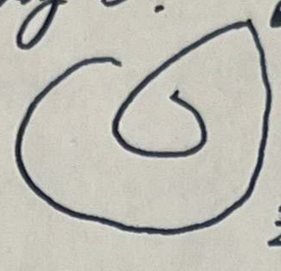


Exa



$\cong S^2 \times S^1$

(;) $[m] \mapsto 0[m] + 1 \cdot [l]$.



$\cong S^3$

$[m] \mapsto 1[m] + 1 \cdot [l]$.

Kirby calculus

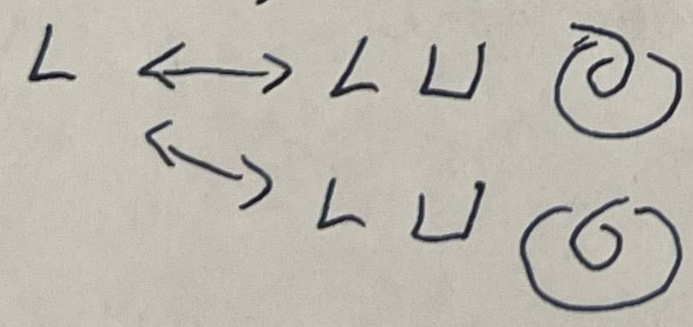
$$\{\text{framed link diagrams}\} \Big/ \Omega_1', \Omega_2, \Omega_3 \longrightarrow \{3\text{-mfd}\} \Big/ \text{homeo.}$$

$$\{\text{framed link diagrams}\} \Big/ \Omega_1', \Omega_2, \Omega_3 \quad (1\text{-}1)$$

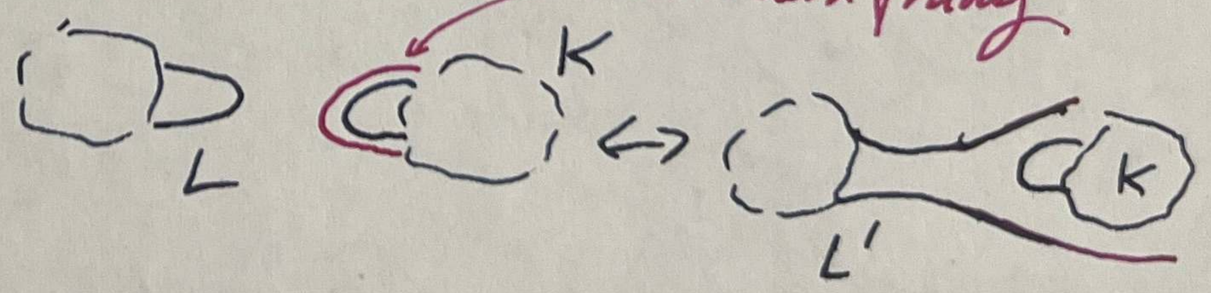
$$+$$

Kirby moves. ???

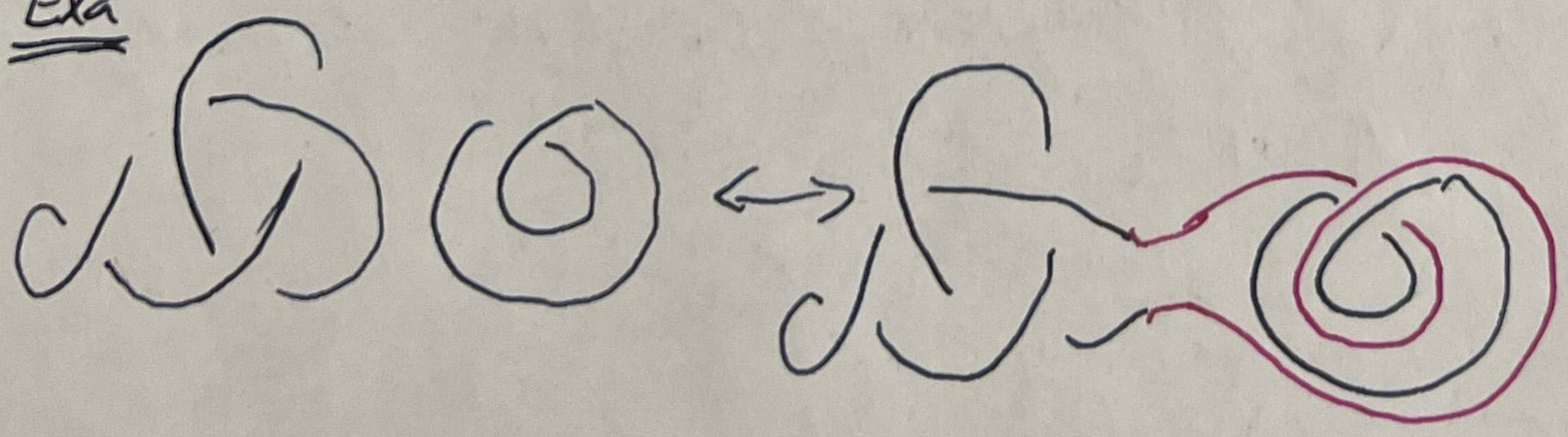
First Kirby move



Second Kirby move



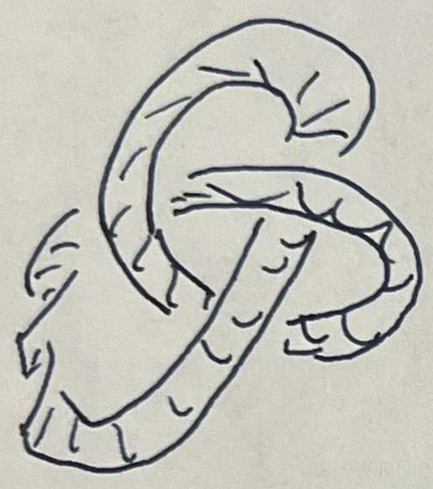
Exa

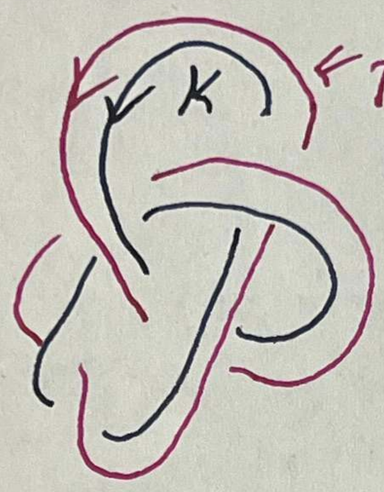


Cor A framed link L determines a 3-manifold obtained from S^3 by Dehn surgery (integer) along $\mathbb{D}L$ w/ homeomorphisms $S^1 \times S^1 \to S^1 \times S^1$ which send meridian to framings of components of L.

Revisit to framed link

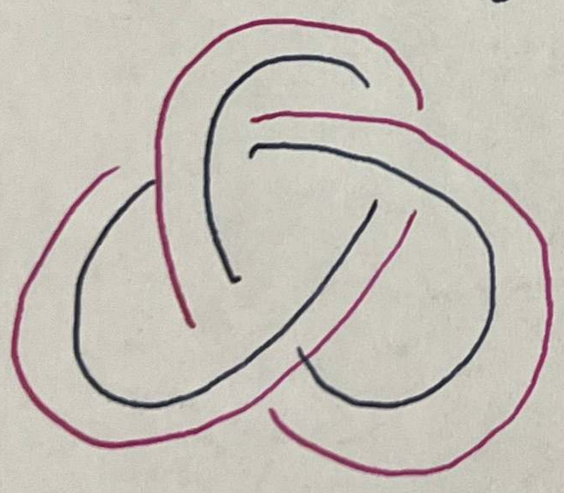
Def A framed knot $S^1 \hookrightarrow S^3$ is a smooth embedding up to ambient isotopy.



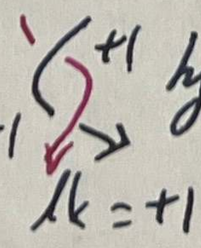


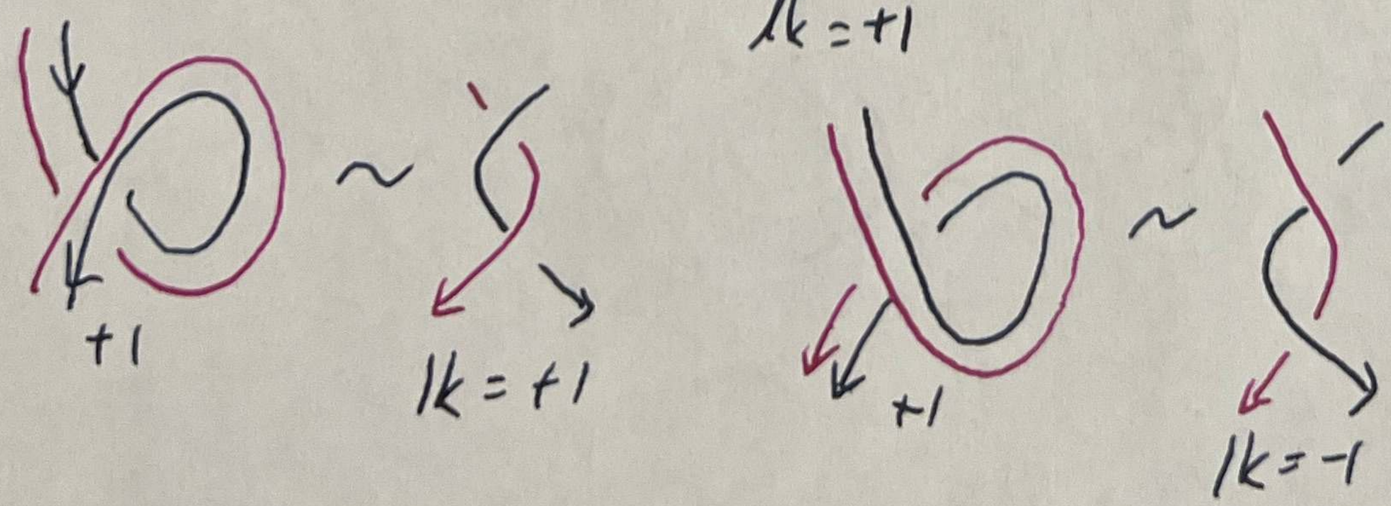
Lemma A framing of K is determined by the linking number between K and K'.

Blackboard framing



The framing of K obtained from blackboard framing of a diagram of K = the writhe of D.

How describe a framing  by diagram w/ blackboard framing?



i.e ∞ {framed links} $\longleftrightarrow$ {diagrams w/ blackboard framing}

Prop Let M be a 3-mfd w/ a torus boundary (exa. $S^3 - (D^2 \times S^1)$)

Let M_ϕ be a 3-mfd obtained by $M_\phi = M \sqcup_\phi D^2 \times S^1$,

$\phi : S^1 \times S^1 \to S^1 \times S^1$.

Assume that $\psi : S^1 \times S^1 \to S^1 \times S^1$ and $\phi(m) = \psi(m)$, where m: meridian.

Then M_ϕ and M_ψ are homeomorphic.

That is, it is enough to determine the image of M to do Dehn surgery along a knot K w/ a homeo : $\phi : S^1 \times S^1 \to S^1 \times S^1$.

Description of images of M by homeo $S^1 \times S^1 \to S^1 \times S^1$

For $\phi : S^1 \times S^1 \to S^1 \times S^1$: a homeo, $\phi(m)$ is a simple closed curve.

$\pi_1(S^1 \times S^1) \cong \mathbb{Z} \oplus \mathbb{Z}$ and generated by $[m]$ and $[l]$, m, l: meridian, l: longitude

$\Rightarrow \phi([m]) = p[m] + q[l]$, where p, q: coprime.

i.e. Image of m by ϕ is determined by rational number $\frac{p}{q}$.

Def For a knot K in S^3, a Dehn surgery along K w/ $\phi : S^1 \times S^1 \to S^1 \times S^1$, $[m] \mapsto p[m] + q[l]$

is called a rational surgery along K.

In particular, if $q = 1$, then it is called an integer surgery.

Thm Every closed orientable 3-manifold can be obtained from the sphere S^3 by integer surgery along a link.

Remark α : a simple closed curve s.t. $[\alpha] = p[m] + 1 \cdot [l]$.

